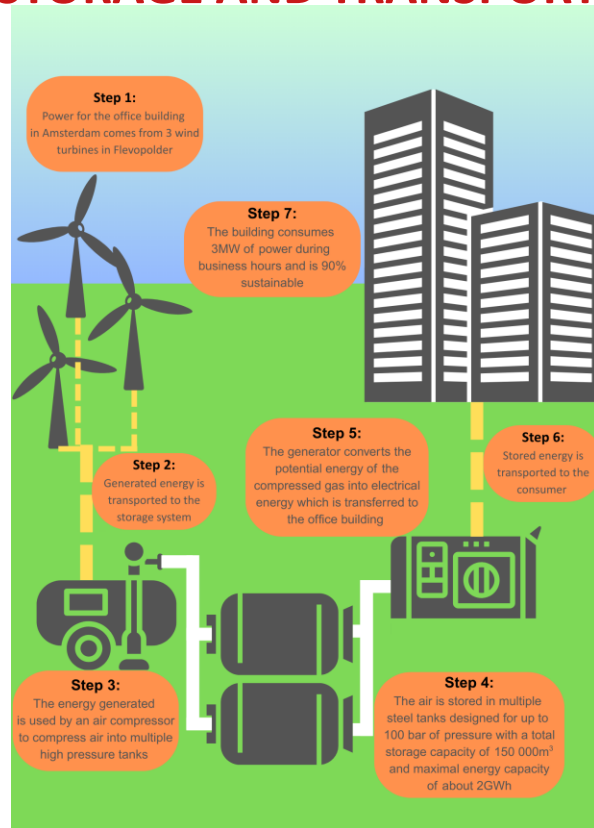


ENERGY STORAGE AND TRANSPORT SYSTEM



MODEL PARAMETERS

```
gamma_val = 1.2; %polytropic constant
P_max = 100e5; % Pa % Max pressure held by tank
tank_volume = 150000; % m^3
buy_price = 0.456; % €/kWh
sell_price = 0.8 * buy_price; % €/kWh
L = 40.0; % m % length of one tank
r = 8.92; % m % radius of one tank
d = 0.7; %m %thickness of one tank
tank_volume = 150000; % m^3 % one tank volume
A = 2740.6; % m^2 % surface are of tank
m = tank_volume/10000; %number of tanks
T0 = 10.6 + 273.15; % K % outside temperature
k = 16.2; % W/(m·K)
p0 = 1e5; % Pa % atmospheric pressure
```

```
% transport from supply
aSupplyTransport = 0.01;
% injection system
aInjection = 0.1;
% storage system
EStorageMax = 500000.*unit("kWh");
EStorageMin = 0.0*unit("kWh");
EStorageInitial =
100000.0*unit("kWh");
bStorage = 0*unit("s");
% extraction system
aExtraction = 0.1;
aDemandTransport = 0.01;
```

```
surplus_power = power_supplied - power_demanded; % -3156276 - 15082639 W
time_diff = [0; diff(time_supplied)]; % 0 - 31535100 s
surplus_energy = surplus_power .* time_diff; % -2.8406e+09 - 1.3574e+10 J
Edot = surplus_power; % -3156276 - 15082639 W
Pdot = ((gamma_val - 1)/ tank_volume)* Edot; % -4.2084 - 20.1102 Pa/s
```

MODEL LAWS AND ASSUMPTIONS

- The basis for the model is **conservation of energy**
- Adjusted **Ideal Gas Law with Compression** is used to estimate air behavior:

$$pV = Z(p, T) \cdot mRT$$

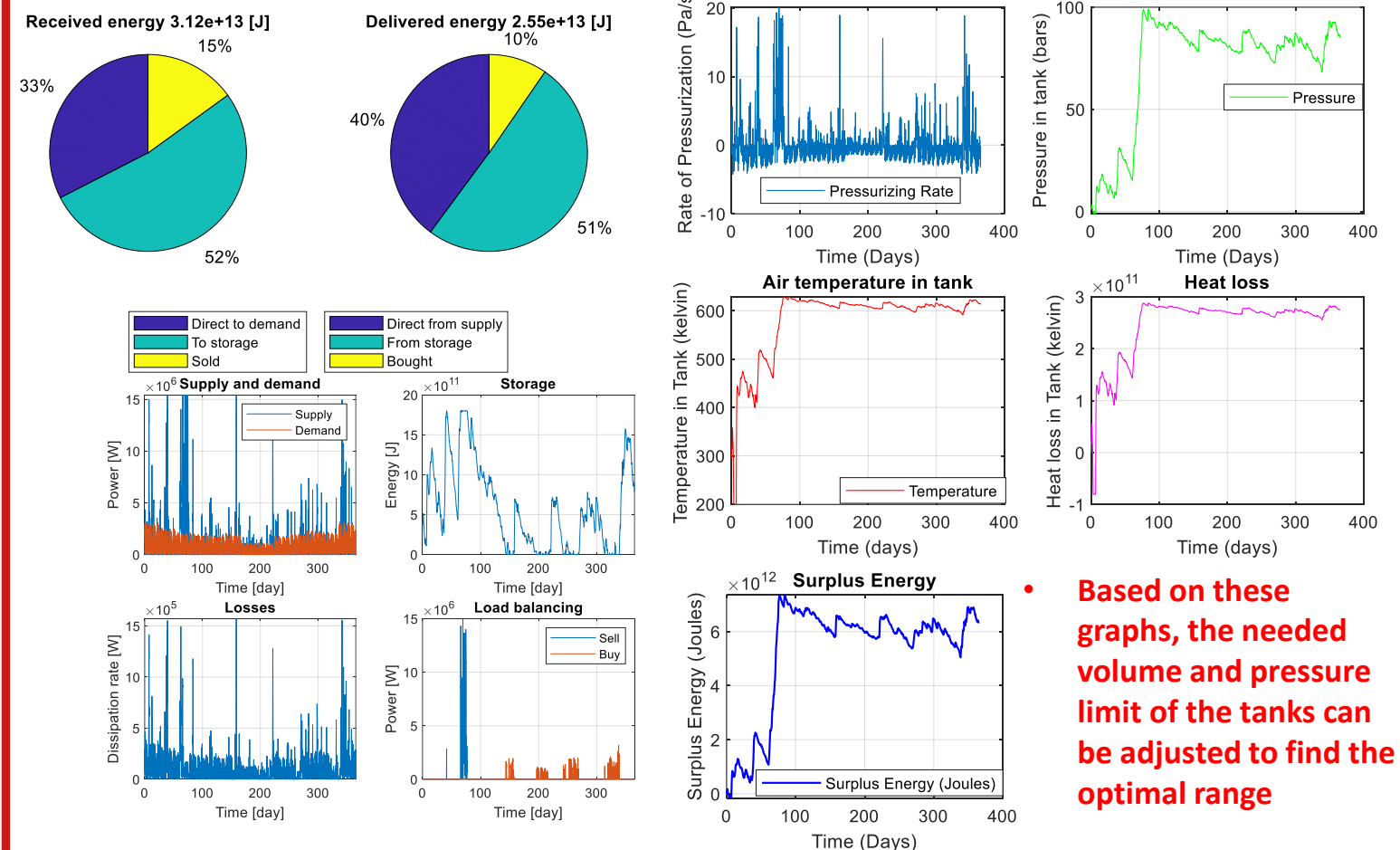
- Since no temperature sensor is provided, **pressure data is used to estimate inside temperature**.
- Outside temperature** is assumed to be **10,6°C**, as that is the average outside temperature.
- Due to the windiness in Flevopolder, **the tank surface temperature is assumed to be in equilibrium with the environment**.
- In real world systems, **compression of air is not perfectly isothermal or adiabatic**.
- Gas compression behavior** in storage tanks is modeled using the **polytropic process**.

$$\frac{dP}{dt} = \frac{(\gamma - 1)}{V} \cdot \dot{E}$$

- Van der Waals Equation** can also predict a more realistic gas behavior, especially at high pressures.

$$\left(p + a \left(\frac{n}{V}\right)^2\right) \cdot (V - nb) = nRT$$

MODEL RESULT



- Based on these graphs, the needed volume and pressure limit of the tanks can be adjusted to find the optimal range